Topological Preliminaries

* A <u>topology</u> T in a set X is a collection of subsets of X (called open sets) satisfying the following axioms:

  (1) the union of any number of open sets is open

  (2) the intersection of any finite number of open sets is open.

  (3) X, ϕ are open.

A <u>topological space</u> is a set X with a topology T in X.

$T = \{\phi, X\}$ is <u>trivial topology</u>

$T = P(X)$ is <u>discrete topology</u>

* For $A \subseteq X$, where T is a topology in X.

Then T_A is an <u>induced topology</u> in A

where $T_A = \{ U \cap A \mid U \in T \}$.

* In a topological spaces, a <u>closed set</u> is a complement of an open set.

* For $E \subseteq X$, $\overline{E}$ is said to be <u>closure</u> of E, where $\overline{E} = \bigcap_{V \in S} V$, S is the set of all closed sets containing E.

* A <u>neighbourhood</u> of $x \in X$, is any open set containing x.

* A <u>mapping</u> f from X_1 to X_2 $(X_1 \xrightarrow{f} X_2)$ is an assignment of each $x \in X_1$ of an element $f(x) \in X_2$ where X_1, X_2 are topological spaces. (map and mapping are not the same, mapping is a function between two topological spaces).

* The mapping f is <u>continuous</u> if and only if for every open set O_2 in X_2, the set $f^{-1}(O_2)$ is an open set in X_1. (note f^{-1} need not be a mapping)

* For $x \in X_1$, we say that f is continuous at x, if to every neighbourhood U of $f(x)$, there exists a nbd V of x, such that $f(V) \subset U$.

(f is continuous at every point of X_1.)

We can show the equivalence of both definitions.

Lemma

Composition of continuous maps are continuous.

Proof: If $f: X \longrightarrow Y$, $g: Y \longrightarrow Z$ are continuous.

Goal 1: $(g \circ f)^{-1}(V)$ is open in X where V is any open set in Z.

Goal 2. $(g \circ f)^{-1}(V) = f^{-1} \circ g^{-1}(V)$

$\square$

Example

(i) Every mapping is continuous from X to Y if X has discrete topology. (I showed converse is NOT true).

(ii) If X is a topological space, Y a set with trivial topology, $f: X \longrightarrow Y$ is continuous.

* <u>Topological product</u> of X_1, X_2 is $X = X_1 \times X_2$ where topology in X is generated by a basis
$\mathcal{B} = \{ O_1 \times O_2 \mid O_1 \in \tau_1, O_2 \in \tau_2 \}$.

(Def^n of basis add to notes ; I'm not writing here)

* If $p_1 : X_1 \times X_2 \longrightarrow X_1$ and
$p_2 : X_1 \times X_2 \longrightarrow X_2$ are such that
$p_1(x_1 \times x_2) = x_1$ and $p_2(x_1 \times x_2) = x_2$ resp.

Then p_1, p_2 are <u>projections</u> of $X_1 \times X_2$.

Lemma

Projections are continuous, open mappings.

Proof: It suffice to show for p_1 and for p_2 is the same argument.

For open map, observe every open set in X can be written as $U \times V$ where U is an open set in X_1 and V is an open set in X_2.

Now, $p_1(U \times V) = U$.

For continunity, observe $p_1^{-1}(U) = U \times X_2$ where $U \underset{\text{open}}{\subset} X_1$.

* A family of subsets $\{V_\alpha\}_{\alpha \in I}$ is a <u>covering</u> if $\bigcup_{\alpha \in I} V_\alpha = X$. If each V_α is an open subset of X, $\{V_\alpha\}_{\alpha \in I}$ is an <u>open covering</u> and if $|I| < +\infty$ then $\{V_\alpha\}_{\alpha \in I}$ is a finite covering.

$\square$

* A topological space X is compact, if every open covering of X contains a finite covering.
A subset A is said to be compact, if A is compact in the induced topology.

Remarks

(i) $\mathbb{R}$ is not compact

(ii) If $A \subseteq X$, and $|A| < +\infty$, then A is compact.

(iii) Let $a, b \in \mathbb{R}$, $a \leq b$ then $[a, b]$ is compact in $\mathbb{R}$.

Proof: refer to theorem 27.1 and corollory 27.2 of Munkres.

Let X be a simply ordered set having least upper bound property. Then every closed interval in X is compact. (Check out prahlad sir's notes on General Topology for alternate path).

Given an open covering of $[a, b]$ say, $\mathcal{A}$.

STEP 1: For $x \in [a, b]$ there exits $y \in [a, b]$ and $y > x$ such that $[x, y]$ is covered by atmost two elements of $\mathcal{A}$. (Do case where imm. succ exist and imm. succ doesn't exist).

STEP 2: $C = \{ y \in [a, b] \mid [a, y]$ are covered by finite elements of $\mathcal{A} \}$

Observe $C \neq \phi$ by applying STEP 1 for $x = a$. and apply least upper bound property where $c = \sup C$.

STEP 3: Show $c \in C$

STEP 4: $b = c$.

* Finite intersection property: In topological space X, let $\{F_\alpha\}_{\alpha \in I}$ be collection of closed sets. If intersection of any finite subcollection of $\{F_\alpha\}_{\alpha\in I}$ is non-empty. Then $\bigcap_{\alpha\in I} F_\alpha \neq \phi$.

A space holds FIT iff the space is cpt.

Theorem

A closed subset of a compact set is compact

Proof: Let X be compact, and Y be a closed subset of X.

For any open covering of Y, say $\mathcal{A}$, we construct $\mathcal{A} \cup \{X \setminus Y\}$ as an open covering of X as $X \setminus Y$ is open. As X is compact, Let $\mathcal{A}'$ be a finite subcover of X. Now, the finite subcover of Y we get from $\mathcal{A}'$ will not contain $\{X \setminus Y\}$, hence finite subcover of $\mathcal{A}$.

Theorem

A continuous image of a compact set is compact.

Proof: ✓

Theorem

If X is compact and Y is hausdorff and $f : X \longrightarrow Y$ is continuous. Then $f(X)$ is closed in Y.

Proof: ✓

* A space X is locally compact, if for every $x \in X$, there is a neighbourhood U such that $\overline{U}$ is compact.

Theorem

Every compact space is locally compact

Proof: ✓

Converse may not be true. Example : $\mathbb{R}$

Proof. ① $\mathbb{R}$ is not compact

② For $x \in \mathbb{R}$, there exists $(x-\varepsilon, x+\varepsilon)$ as a nbd of x and $\overline{(x-\varepsilon, x+\varepsilon)} = [x-\varepsilon, x+\varepsilon]$ is compact.

* Homeomorphism is a mapping between two topological spaces X, Y say f such that f is bijective map where f, f^{-1} are continuous mappings.

Examples

Add from C. chandrashekharan.

Note: All topological spaces are T_1 moving forward unless mentioned otherwise.

* Two sets M_1 and M_2 are <u>seperated by open sets</u> if there exist open sets O_1 and O_2 containing M_1 and M_2 such that $O_1 \cap O_2 = \emptyset$.

* Two sets M_1 and M_2 are seperated by a <u>real function</u> if there exists a continuous real function $f: X \longrightarrow [0, 1]$ such that $f(x) = 0$ for $x \in M_1$ and $f(x) = 1$ for $x \in M_2$.

T_1: For any disjoint point, there exists an open set containing one and not the other. Equivalently, all singleton are closed.

T_2 (Hausdorff): Two distinct points can be seperated by open set

T_3 (Regular): For closed set F and $x \notin F$, can be seperated by open sets

T_4 (Normal): Disjoint closed sets can be seperated by open sets.

Remark : $T_4 \Rightarrow T_3 \Rightarrow T_2 \Rightarrow T_1$.

Examples : From the book.

Urysohn's lemma

In a normal space, every two disjoint closed sets can be seperated by a real function.

Tychanoff Theorem

Product of compact sets are compact

There are two approaches of proving this theorem

1. Zorn's lemma
2. Transfinite Induction